

MERIDIAN

Log ingestion, normalisation, retention and rotation for hybrid estates

How on-premises, cloud and SaaS logs reach MERIDIAN: six ingestion paths, from cloud-native exports and syslog over TLS to Windows Event Forwarding with NXLog, Sysmon, log shippers, HTTPS push and API pull. How every record is normalised to one OCSF schema without losing anything, how long each kind of log is kept and how it is rotated, and how to onboard, size and operate each source.

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Getting the logs in

Most SIEM projects are decided by ingestion: what can be collected, what it costs, whether it parses, and how long it is kept. This document is MERIDIAN's answer, path by path.

WHAT THIS DOCUMENT IS, AND HOW IT RELATES TO THE OTHERS

This is the design and build document for MERIDIAN's ingestion tier in a hybrid estate: Windows and Linux servers, network and OT devices on premises, Azure and AWS platforms, and SaaS applications.

It covers six things: the ingestion paths and when to use each; the Windows and Sysmon design; normalisation; retention and rotation; sizing and operations; and the onboarding procedure.

The platform build is in AZ-00 (Azure) and AWS-00 (AWS); this document adds only what ingestion needs. Where it and the code disagree, the code (`meridian/ingest`, `meridian/mappers`) and Terraform are the truth.

1 Why ingestion is where SIEM projects struggle

The same seven problems appear in almost every SIEM programme. MERIDIAN's design starts from them.

Problem with a conventional SIEM	What it causes	What MERIDIAN does instead
Licence priced per GB ingested	Teams filter out firewall, DNS, flow and endpoint logs to stay under the cap. The data needed in an investigation was never collected	Storage is billed at object-storage prices, with no ingestion licence and no daily cap. Collect everything that has security value; filter for noise, not cost
Parsing at the edge, before storage	A vendor changes its log format, the parser fails and events are dropped or mis-filed until someone notices	Land raw first, parse later. Every batch is stored exactly as received for 90 days. A mapper fix is followed by a replay, so nothing is lost
A different schema per source and per tool	Rules and queries are written per vendor; migrations rewrite them	One schema: OCSF classes in 42 typed columns. The same columns work in DuckDB, Athena and Azure Data Explorer, and for every agent and rule
Agent sprawl	A proprietary agent on every server, with its own upgrade cycle	No MERIDIAN agent. Native forwarding (WEF, diagnostic settings, CloudTrail), standard syslog, and any shipper you already run
Retention tied to the licence or to hot storage	A year of logs is unaffordable; old logs go to an archive that cannot be searched	Retention is a storage setting per log class, enforced by WORM (Object Lock or immutability). Older data moves to cheaper tiers and stays queryable
Silent sources go unnoticed	A source stops sending and nobody knows until an incident	Per-source freshness, event counts and reject counts are metrics with alerts; LODESTAR reports silent sources as a KRI
Lock-in through formats and forwarders	Leaving the SIEM means re-plumbing every source	Open formats (Parquet, OCSF), standard protocols, and a Splunk HEC-compatible endpoint so existing senders can be re-pointed

2 Architecture



Figure 2.1 Six ways in, one landing zone, one schema. Everything is private and authenticated.

Design principles

- 1. Land raw, then normalise.** Collection never depends on parsing. The landing zone holds each batch exactly as received, compressed and immutable, for 90 days, and the lake can be rebuilt from it.
- 2. One schema.** Every record becomes one or more OCSF-flat events. The full original record is kept in the raw column for forensics.
- 3. Never drop silently.** A record that no mapping recognises is kept as an OCSF Base Event (class 0) with its host, program and message. A record that cannot be read at all is counted as a reject, and rejects are a metric.
- 4. Every path is private and authenticated,** and each sender can write only into its own source prefix.
- 5. Retention is policy, not price.** WORM for the minimum period, tiering for cost, and a per-class end of life.
- 6. No model in the ingestion path.** Ingestion and first-line detection are deterministic, cheap and auditable. Agents see alerts and query results, never the firehose.

Components

Component	Role	Runs as
Landing zone	Raw batches <source>/<yyyy>/<mm>/<dd>/<batch>.<ext>, gzip, 90-day expiry	ADLS Gen2 container / S3 bucket (SSE-KMS)

Component	Role	Runs as
Queue	New-batch notification, at-least-once delivery	Event Grid → Storage queue / S3 event → SQS with a dead-letter queue
Ingestion worker	Mapper → enrichment → de-duplication → Parquet → streaming detection	<code>meridian worker</code> ; scales on queue depth (KEDA / ECS autoscaling)
Syslog receiver	UDP, TCP and TLS (RFC 5425) listener, buffered batches to landing	<code>meridian syslog</code> (ECS behind your NLB; Container App on Azure)
Push endpoint	<code>/api/ingest/<source></code> (HMAC or token) and <code>/services/collector</code> (HEC)	The API service, internal only
API collector	Polls SaaS and management APIs, with cursors	<code>meridian collect</code> : one active replica, lease-locked
Lake	OCSF-flat Parquet events/ <code>cls=/dt=/hr=</code> , WORM, tiered	ADLS / S3; queried by DuckDB, Athena or Azure Data Explorer

3 Choosing an ingestion path

The six paths

Path	Best for	Authentication	Latency
1 Cloud-native export to object storage	Azure and AWS platform logs; anything a vendor can export to Blob or S3	Cloud IAM, inside your tenancy	Minutes (the provider's delivery interval)
2 Streaming capture	High-volume Azure feeds (Event Hubs Capture) and AWS streams (Firehose)	Cloud IAM	1–5 minutes (capture window)
3 Syslog receiver	Network, security and OT devices, Linux, NXLog CE, log relays	Network allow-list; TLS with optional client certificates	≤ 30 seconds (flush interval)
4 HTTPS push	Log shippers, custom applications, existing Splunk HEC senders	HMAC signature or a token bound to one source	Seconds
5 API pull	SaaS audit logs exposed only through APIs	OAuth2 client credentials, bearer or basic, held in the vault	Interval (default 5 minutes) plus the API's own delay
6 Shipper to object storage	Kubernetes, large Linux estates, Cribl or Vector pipelines	IAM role limited to one source prefix (AWS); managed identity (Azure)	Shipper batch interval

Source catalogue

Source	Recommended path	Alternatives	Mapper	Notes
Windows servers and domain controllers: Security, System, PowerShell	Windows Event Forwarding to a collector (WEC), NXLog CE om_ssl → TLS syslog (3)	Fluent Bit winevtlog → push (4); NXLog Agent om_http → push; Winlogbeat → Logstash → push or S3	windows (through syslog)	Section 5. Filter to the event IDs you use at the collector
Sysmon	Same as Windows (Sysmon channel in the WEF subscription)	Same	windows	Use a maintained Sysmon configuration; image-load and registry events are the volume drivers
Windows endpoints with Defender for Endpoint	Defender XDR streaming API → Event Hubs or Storage (1 / 2)	–	mde	No Windows forwarding needed for these hosts
Linux servers	rsyslog or syslog-ng with TLS → syslog (3)	Fluent Bit or Vector (journald, files) → S3/Blob (6) or push (4)	syslog	sshd, sudo and useradd are recognised; auditd through its syslog plugin
Network devices (switches, routers, wireless)	UDP syslog to a local relay (rsyslog), relay → TLS syslog (3)	Direct UDP/TCP to the AWS receiver	syslog + patterns	Add a pattern per message you want as a typed event; the rest is kept as Base Events

Source	Recommended path	Alternatives	Mapper	Notes
Firewalls, proxies, WAF, VPN, IPS	CEF or LEEF over syslog (3)	Vendor cloud log export to S3/Blob (1)	syslog (CEF/LEEF auto) or cef	Palo Alto, Fortinet, Check Point, F5, Zscaler NSS, Netskope
OT, port and terminal systems, appliances	CEF/LEEF syslog to a relay in the OT DMZ, relay → TLS syslog (3)	–	syslog	UDP syslog crosses one-way gateways (data diodes); TCP and TLS do not
Azure: Entra ID, Activity log, resource logs, Defender for Cloud	Diagnostic settings or continuous export → Event Hubs → Capture, or → Storage (1 / 2)	–	entra, json (field map)	VNet flow logs to Storage; NSG flow logs retire on 30 September 2027
AWS: CloudTrail, VPC Flow Logs, Route 53 Resolver, GuardDuty, ALB/WAF	Organisation trail and service exports to S3 (1)	Security Lake subscriber (OCSF)	cloudtrail, ocsf, json	Partition by account and region; the landing prefix maps to the source
AWS CloudWatch Logs (applications, Lambda, EKS)	Subscription filter → Firehose → S3 (2)	–	json / syslog	
Kubernetes and containers	Fluent Bit DaemonSet → S3/Blob (6) or push (4)	Vector	json / syslog	
Microsoft 365 audit	API pull, Management Activity API (5)	Entra and Defender through Event Hubs (1)	json (field map)	Two-step API; start the subscription once
Okta, GitHub, Google Workspace, Atlassian, Salesforce	API pull (5)	Vendor streaming to S3/Blob where offered (1)	json (field map)	Link, next-field and header pagination are supported
CrowdStrike, Zscaler and other OCSF producers	Vendor export or Security Lake (1)	Push (4)	ocsf	
Existing SIEM during migration	Splunk HEC senders → /services/collector (4); QRadar/ArcSight forwarders → syslog LEEF/CEF (3)	Log Analytics data export → Storage / Event Hubs (1)	per source	Run side by side, then cut over source by source
Custom applications	HTTPS push with HMAC (4)	JSON lines into landing (6)	json (field map)	scripts/push_events.py signs requests for tests

DO NOT BUILD ON AZURE MONITOR AGENT "DIRECT TO STORE"

The preview that let Azure Monitor Agent send Windows events and syslog straight to Event Hubs or Storage is being retired. New data collection rules could not be created from February 2026, and existing ones stop sending after **31 July 2026**. Microsoft's replacement routes data through Log Analytics first, which brings back per-GB ingestion charges. For Windows and Linux on Azure, use the paths in this document (WEF with NXLog, Fluent Bit, rsyslog), not AMA.

4 The six paths in detail

4.1 Cloud-native export to object storage

The provider writes files into the landing container or bucket. A storage event puts a message on the queue, and the worker processes the batch.

- **Mapping.** A source is matched by the longest prefix: `{key: cloudtrail, format: cloudtrail, prefix: AWSLogs}`. Envelopes such as Azure `{"records": [...]}`, CloudTrail `{"Records": [...]}` and Graph `{"value": [...]}` are unwrapped automatically.
- **Formats.** JSON, JSON lines and text lines, each optionally gzip-compressed, plus Event Hubs Capture Avro.
- **Cross-account (AWS).** Point or replicate the organisation trail into the landing bucket. Set `cloudtrail_account_ids` so that both the bucket policy and the KMS key policy allow it.

4.2 Streaming capture

For feeds that only stream (Defender XDR, Entra, Activity log), send them to an Event Hub with **Capture** to the landing container. Capture writes Avro every 1–5 minutes or every N MB, and MERIDIAN reads the Body of each record. On AWS, Kinesis Data Firehose writes gzip JSON to the landing bucket with a buffering hint (for example 5 MB / 60 s).

4.3 Syslog receiver

Property	Value
Listeners	<code>--port</code> UDP + TCP (default 5514); <code>--tls-port</code> TLS (6514 by convention); either can be off (0)
Framing	Newline-delimited or octet-counted (RFC 6587 / RFC 5425)
Header formats	RFC 5424; RFC 3164 with or without <PRI>; ISO-timestamp forwarding formats
Payloads	CEF, LEEF 1.0/2.0, JSON (Windows events from NXLog, or generic), plain text
TLS	TLS 1.2 or later; certificate and key as files or as PEM in <code>MERIDIAN_SYSLOG_TLS_CERT</code> / <code>_KEY</code> (held in memory, never written to disk)
Mutual TLS	<code>--tls-client-ca</code> : only senders with a certificate from that CA can connect
Buffering	Flushes every 30 s or 20,000 lines. A storage outage keeps lines in memory (up to 500,000) and retries; lines are capped at 64 KB
AWS deployment	ECS service, 2 tasks, UDP and TCP 5514 behind your Network Load Balancer. For TLS, use an NLB TLS listener with an ACM certificate in front of TCP 5514, or run with <code>--tls-port</code>
Azure deployment	Container App (<code>syslog_receiver = true</code>), TCP 6514 with TLS on the environment's internal IP. Container Apps has no UDP ingress: UDP senders go through a relay

A relay per site is the recommended pattern for UDP-only devices. rsyslog or syslog-ng on a small VM receives UDP locally and forwards over TLS with a disk queue, so a WAN outage delays data instead of losing it.

rsyslog, forwarding to MERIDIAN over TLS with a client certificate and a disk-assisted queue:

```
global(DefaultNetstreamDriver="gtls"
  DefaultNetstreamDriverCAFile="/etc/pki/meridian/ca.pem"
  DefaultNetstreamDriverCertFile="/etc/pki/meridian/client.pem"
  DefaultNetstreamDriverKeyFile="/etc/pki/meridian/client.key")
action(type="omfwd" target="syslog.meridian.internal.example" port="6514" protocol="tcp"
  StreamDriver="gtls" StreamDriverMode="1" StreamDriverAuthMode="x509/name"
  StreamDriverPermittedPeers="syslog.meridian.internal.example"
  TCP_Framing="octet-counted" template="RSYSLOG_SyslogProtocol23Format"
  queue.type="LinkedList" queue.filename="meridian_fwd" queue.maxDiskSpace="2g"
  queue.saveOnShutdown="on" action.resumeRetryCount="-1")
```

syslog-ng:

```
destination d_meridian {
  syslog("syslog.meridian.internal.example" transport("tls") port(6514)
    tls(ca-file("/etc/pki/meridian/ca.pem") cert-file("/etc/pki/meridian/client.pem")
      key-file("/etc/pki/meridian/client.key") peer-verify(required-trusted))
    disk-buffer(disk-buf-size(2000000000) reliable(yes)));
};
```

4.4 HTTPS push

Endpoint	Authentication	Body
POST <code>/api/ingest/<source></code>	HMAC: X-Meridian-Timestamp + X-Meridian-Signature: sha256=HMAC(secret, "<ts>.<body>"), 5-minute window; or Authorization: Bearer <token> bound to <source>	JSON object or array, <code>{"records": [...]}</code> , NDJSON, or text lines; Content-Encoding: gzip
POST <code>/services/collector[/event]</code>	Authorization: Splunk <token> (or Bearer); the token decides the source	Splunk HEC events (concatenated <code>{"event": ...}</code> objects)
POST <code>/services/collector/raw</code>	Same	Raw lines
GET <code>/services/collector/health</code>	None	HEC health check

- **Limits.** 10 MB per request (compressed), 64 MB once decompressed (413 above it), and 100,000 records per request.
- **Tokens.** MERIDIAN_INGEST_TOKENS = "source:token, ..." (secret ingest-tokens; none disables tokens). Each token is at least 32 characters and is valid for one source only.
- **HEC.** Indexer acknowledgement is not offered. Senders that require it must turn it off.

Fluent Bit on Windows, pushing Security, System, Sysmon and PowerShell events:

```
[INPUT]
  Name      winevtlog
  Channels  Security, System, Microsoft-Windows-Sysmon/Operational, Microsoft-Windows-
PowerShell/Operational
  Interval_Sec  1
  Event_Data_As_Map true
  DB            C:\fluent-bit\winevtlog.sqlite

[OUTPUT]
  Name      http
  Match     *
  Host      meridian.internal.example
  Port      443
  URI       /api/ingest/windows
  Format     json_lines
  Header     Authorization Bearer ${MERIDIAN_INGEST_TOKEN}
  Compress   gzip
  tls        On
  tls.verify On
  tls.ca_file C:\fluent-bit\ca.pem
```

Event_Data_As_Map true is required: without it Fluent Bit sends positional StringInserts, which change between Sysmon schema versions and are not mapped.

4.5 API pull collector

The collector reads pull: blocks in the source configuration. It runs as its own role (meridian collect, one active replica, ingestion identity) and writes what it fetched to the landing zone.

Option	Values
auth	bearer {scheme, token}, header {name, token}, oauth2 {endpoint, client_id, client_secret, scope}, basic {username, password}, none. Secrets must be \${ENV} references (literal secrets fail at load)
params	Query parameters with {cursor}, {end} and {now} placeholders
records	Dotted path to the list in the response (" ", value, items ...)
paginate	link (RFC 8288 Link: rel="next"), next_field {path} (for example @odata.nextLink), header {name} (for example NextPageUri), none
expand	Field holding a URL to fetch per item: two-step APIs such as Microsoft 365 contentUri
cursor	field (record time), mode max or window, start_minutes, max_window_minutes, format (strftime)
interval_minutes, max_pages	Defaults 5 and 50
Secrets	extra_secrets = ["okta-token"] in Terraform creates the vault entry and exposes MERIDIAN_OKTA_TOKEN

The cursor advances only after the batches are in the landing zone. HTTP 429 or 5xx ends the run without moving the cursor, and the next run resumes from the same point.

Microsoft 365 Management Activity API (start the subscription once with /subscriptions/start):

```
- key: m365
format: json
settings: {class_uid: 6003, field_map: {time: CreationTime, user: UserId, api_operation: Operation,
    src_ip: ClientIP, app_name: Workload, resource: ObjectId, status: ResultStatus}}
pull:
  url: https://manage.office.com/api/v1.0/<tenant-id>/activity/feed/subscriptions/content
  auth: {type: oauth2, endpoint: "https://login.microsoftonline.com/<tenant-id>/oauth2/v2.0/token",
    client_id: <app id>, client_secret: "${MERIDIAN_0365_CLIENT_SECRET}",
    scope: "https://manage.office.com/.default"}
  params: {contentType: Audit.General, startTime: "{cursor}", endTime: "{end}"}
  paginate: {type: header, name: NextPageUri}
  expand: contentUri
  cursor: {mode: window, start_minutes: 60, max_window_minutes: 1440, format: "%Y-%m-%dT%H:%M:%S"}
  interval_minutes: 5
```

The window is capped at 24 hours because the API refuses longer ranges. It also refuses start times more than 7 days back. If the collector is down for longer than that, the gap cannot be recovered from this API.

4.6 Shipper straight to object storage

Fluent Bit, Vector, Cribl, Logstash or NXLog Agent write batches to <source>/... in the landing zone.

- **AWS.** Set `landing_writer_sources = ["k8s", "linux"]`. Terraform creates one managed policy per source, allowing `s3:PutObject` under that prefix only, with the lake KMS key. Attach it to the shipper's role: an EKS service account, an EC2 instance role, or IAM Roles Anywhere for on-premises collectors.
- **Azure.** The lake account is keyless and private. In-Azure shippers use a managed identity with Storage Blob Data Contributor on the landing container, over the private endpoint. On-premises shippers should use push (4.4) or syslog (4.3) instead, because shared keys are disabled.

Vector on Linux, journald to the landing bucket:

```
[sources.journal]
type = "journald"

[sinks.meridian]
type = "aws_s3"
inputs = ["journal"]
bucket = "<landing bucket>"
key_prefix = "linux/%Y/%m/%d/"
region = "me-central-1"
compression = "gzip"
encoding.codec = "json"
framing.method = "newline_delimited"
server_side_encryption = "aws:kms"
ssekms_key_id = "<lake KMS key ARN>"
```

With `{key: linux, format: syslog}`, journald records are read with their host (`_HOSTNAME`), program (`SYSLOG_IDENTIFIER`) and message.

5 Windows and Sysmon

Windows is the largest and most valuable on-premises source, and the one most often collected badly. The design avoids an agent on every server.

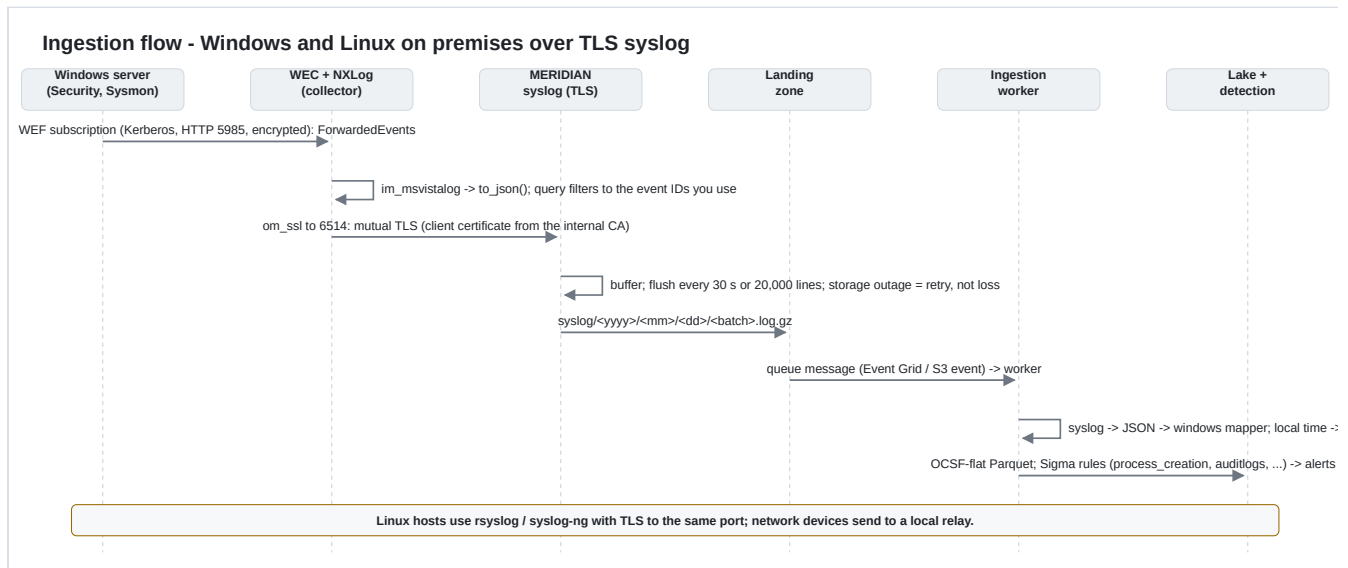


Figure 5.1 Windows Event Forwarding to a collector, NXLog CE to MERIDIAN's TLS syslog receiver.

Collection design

- **Windows Event Forwarding (WEF).** Built into Windows. Servers send to one or more Windows Event Collectors (WEC) under a source-initiated subscription set by Group Policy. Transport is WinRM (HTTP 5985), encrypted with Kerberos inside the domain. Use one WEC per site or per 2,000–4,000 sources, and two for resilience.
- **On the WEC.** NXLog CE (`im_msvistalog on ForwardedEvents → to_json() → om_ssl to 6514` with a client certificate). Fluent Bit (`winevtlog on ForwardedEvents → push`) is the alternative.
- **Hosts that cannot use WEF** (workgroup servers, DMZ) run NXLog or Fluent Bit locally with the same output.
- **Time zone.** NXLog writes `EventTime` in local time without a zone. Set `timezone` on the source (for example `+04:00` or `Asia/Dubai`). Sysmon's `UtcTime` is always used when present.

NXLog CE on the collector:

```

<Extension json>
  Module xm_json
</Extension>
<Input wec>
  Module im_msvistalog
  <QueryXML>
    <QueryList><Query Id="0"><Select Path="ForwardedEvents">*</Select></Query></QueryList>
  </QueryXML>
  Exec to_json();
</Input>
<Output meridian>
  Module om_ssl
  Host syslog.meridian.internal.example
  Port 6514
  CAFile C:\Program Files\nxlog\cert\ca.pem
  CertFile C:\Program Files\nxlog\cert\wec01.pem
  CertKeyFile C:\Program Files\nxlog\cert\wec01.key
</Output>
<Route wec_to_meridian>
  Path wec => meridian
</Route>

```

Audit policy prerequisites

Setting (Group Policy, Advanced Audit Policy)	Why
Logon/Logoff: Logon, Account Lockout, Special Logon — success and failure	4624, 4625, 4740, 4672
Account Logon: Kerberos Authentication Service, Kerberos Service Ticket Operations, Credential Validation (on DCs)	4768, 4769, 4771, 4776: spraying, Kerberoasting, NTLM use
Account Management: User Account Management, Security Group Management	4720–4738, 4728/4732/4756
Detailed Tracking: Process Creation, plus "Include command line in process creation events"	4688 with the command line, for hosts without Sysmon or an EDR
Object Access: Other Object Access Events	4698 (scheduled task created)
PowerShell: Script Block Logging	4104
Security log size and retention	Large enough to cover a WEC outage (for example 1 GB on DCs)

What is mapped

Channel	Event IDs	OCSF class	Used by
Security	4624, 4625, 4634, 4647, 4648, 4768, 4769, 4771, 4776	3002 Authentication	Spray, brute force, sign-in correlations
Security	4672	3005 User Access Management	Hunting
Security	4688	1007 Process Activity	All process_creation Sigma rules
Security	4720, 4722–4726, 4738, 4740, 4767, 4781	3001 Account Change	Account and lockout rules

Channel	Event IDs	OCSF class	Used by
Security	4728, 4729, 4732, 4733, 4756, 4757	3005 User Access Management	mer-id-ad-priv-group-add
Security / System	4697, 4698, 4702, 7045	6003 API Activity (service, task)	Persistence hunting
Security / System	1102, 104	6003 (log cleared, high severity)	mer-ep-event-log-cleared
Sysmon	1, 5, 8, 10, 25	1007 Process Activity	process_creation rules; LSASS access (10)
Sysmon	3 / 22	4001 Network / 4003 DNS	IOC matching, network rules
Sysmon	11, 15, 23, 26	1001 File System Activity	Ransom-note and file rules
Sysmon	12–14 (registry) and anything else	0 Base Event (kept, searchable)	Hunting
PowerShell	4104	1007 (script block in message)	Hunting
Defender Antivirus	1006, 1116, 1117	2004 Detection Finding	Vendor-detection rule

PROVEN IN THE DEMO

The demo estate includes a Windows file server without an EDR. Its Security log arrives as NXLog JSON in local time, and its Sysmon log as Winlogbeat JSON. The same run includes a Linux jump host over RFC 3164 syslog. From those logs MERIDIAN raises:

- brute force (30 × 4625);
- an account added to Administrators (4732);
- a certutil download (Sysmon 1);
- the threat-intelligence domain (Sysmon 22);
- the Security log being cleared (1102).

The run rejects nothing.

6 Normalisation

The schema

Every event is OCSF-flat: 42 typed columns shared by all classes (time, class, activity, severity, status, user, source and destination IP and port, device, process, file, URL, DNS, action, cloud account, API operation, resource, message and others). Two columns are special:

- `raw` keeps the original record;
- `event_uid` is a stable hash used for de-duplication.

Columns are an allow-list: rules, queries and MCP tools can reference only these names.

Mappers

format	Input	Produces
syslog	RFC 5424/3164 lines, or structured records (journald, Vector, Fluent Bit); routes CEF, LEEF and Windows JSON; built-in patterns (sshd, sudo, useradd, usermod) plus your patterns	Typed classes, or 0 Base Event
cef	ArcSight CEF (alone or inside syslog)	4001 / 4002 / 4003 / 3002
leef	IBM LEEF 1.0 and 2.0 (custom delimiters)	4001 / 4002 / 4003 / 3002
windows	NXLog, Winlogbeat, Fluent Bit (map), rendered XML, XML-to-JSON	Section 5 table
mde	Defender XDR advanced-hunting stream	1007, 4001/4002, 1001, 3002, 2004
entra	Entra ID sign-in and audit logs	3002, 3001, 3005, 6003
cloudtrail	AWS CloudTrail	6003, 3002
ocsf	Any OCSF producer (Security Lake, CrowdStrike, Okta, Zscaler)	Flattened as-is
json	Anything else, with <code>class_uid</code> (or <code>class_field</code> + <code>class_map</code>) and a <code>field_map</code> of dotted paths	Configured class

Rules applied to every event

- **Time.** UTC. Naive timestamps take the source's timezone. Fractional seconds and offsets are honoured. RFC 3164 times get the year inferred, so December logs read in January land in the right year.
- **Identity.** Users and hosts in lower case. DOMAIN\user is split into user and user_domain. IPv4-mapped IPv6 (::ffff:) is reduced to IPv4. Image paths are reduced to the file name in process_name, with the full path in file_path.
- **Windows placeholders.** Values such as - become null.
- **Enrichment.** Asset criticality and crown-jewel tags from the CMDB, identity attributes, and threat-intelligence indicator hits (ioc_hits), all at ingestion.
- **De-duplication.** Within a batch by event_uid. Re-delivered batches are skipped by the processed-batch ledger. Replays are idempotent.

Previewing a source before it goes live

```
meridian map-test --file sample.jsonl --format windows --timezone +04:00
format=windows records=1200 events=1200 rejected=0
  class 3002 Authentication: 840
  class 1007 Process Activity: 310
  class 0 Base Event: 50
  columns populated: time(1200), device(1200), user(1150), src_ip(820), process_name(310), ...
```

map-test stores nothing. Run it on a sample from every new source, and keep the output with the onboarding record.

Adding a mapper

A mapper is one function (record, settings) -> [events] registered in meridian/mappers/__init__.py. It must be deterministic and side-effect free, build events with `ocsf.event()` (which rejects unknown columns), and come with tests on real samples. Field maps and patterns cover most new sources without code.

7 Retention and rotation

Layers

Layer	Holds	Kept	Controlled by
Source	Windows event logs, syslog relay disk queues, shipper buffers	Size-based (the operating system or shipper)	Your build standards: big enough to ride out a WAN or collector outage
Landing zone	Raw batches as received	90 days , then deleted	Lifecycle rule (both clouds). Cool tier after 7 days on Azure
Lake	OCSF Parquet per class, day and hour	WORM for lake_retention_days (default 365)	S3 Object Lock (GOVERNANCE or COMPLIANCE) / Azure immutability policy
Lake tiering	Same	Azure: cool at 30 d, cold at 90 d, archive at 400 d; AWS: Standard-IA at 30 d, Glacier Instant Retrieval at 180 d	Lifecycle rules
Lake end of life	Same	lake_expire_days, or per class lake_expire_days_by_class; 0 = keep	Lifecycle rule per class; never earlier than WORM (validated)
Azure Data Explorer (medium and large tiers)	Hot copy for fast hunting	Hot cache 31 days, retention 400 days	Table policies
Operational store	Alerts, cases, approvals, agent runs, audit chain	Kept	PostgreSQL with 35-day backups
Processed-batch ledger, finished work	Bookkeeping	120 and 30 days (store.ledger_days, store.work_days)	Daily housekeeping in the scheduler
Platform logs	MERIDIAN's own logs	30 days	CloudWatch Logs / Log Analytics (with a daily cap)

Example policy

```
lake_retention_days      = 400           # WORM minimum: every log for 13 months
lake_expire_days        = 400           # default end of life
lake_expire_days_by_class = {
  "3002" = 2555                         # authentication: 7 years
  "3001" = 2555                         # account changes: 7 years
  "6003" = 2555                         # cloud and API audit: 7 years
  "4001" = 400                          # network flows: 13 months
}
```

This example meets, for instance, PCI DSS v4.0 requirement 10.5.1: at least 12 months of audit log history, with the latest three months immediately available. MERIDIAN keeps all of it queryable. Set the figures from your own regulatory and legal register; MERIDIAN only enforces them.

DECISIONS THAT ARE HARD TO UNDO

- **COMPLIANCE-mode Object Lock** (AWS) and a **locked immutability policy** (Azure) cannot be shortened by anyone for the retention period. Take legal advice first. GOVERNANCE mode and unlocked policies are the defaults.
- **Shortening the end of life** deletes data at the next lifecycle run. Treat a change to `lake_expire_days*` as a change to a records-retention policy, with approval.

Rotation of everything else

- **Keys.** The KMS key (AWS) and Key Vault keys rotate automatically.
- **Secrets.** Ingest secrets, source tokens and pull credentials are rotated by writing a new value and restarting the receiving role. A token is bound to one source, so rotating one sender does not affect the others.
- **Syslog certificates.** Issue from your internal CA with a lifetime you can renew automatically. With mutual TLS, revoke a collector by removing its certificate from the trusted CA bundle.
- **Batches.** Landing and lake objects are written once and never modified, so there is nothing to rotate in place.

8 Sizing

Measure	Value	How it was obtained
Mapper throughput (one core)	Windows/Sysmon about 40,000 events/s; syslog with patterns about 43,000; CEF about 14,000	Measured in the build sandbox, mapper only
Worker throughput, end to end (one process)	About 9,600 events/s: map, enrich, de-duplicate, write Parquet, evaluate 30 streaming rules	Measured on 100,000 Sysmon events in the build sandbox
200 GB/day at about 500 bytes per event	About 4,600 events/s on average	Arithmetic
Worker replicas	AWS: 2 to 20 on queue depth; Azure: 1 to 10 (KEDA rule on the landing queue)	Terraform defaults
Syslog receiver	2 tasks (AWS) or 2–4 replicas (Azure); one receiver per source key	Terraform defaults

The sandbox figures are a floor, not a promise: measure your own mix during the shadow run (AZ-00 / AWS-00 stage S6). Raise `max_events_per_batch` only if producers cannot split batches.

9 Operations

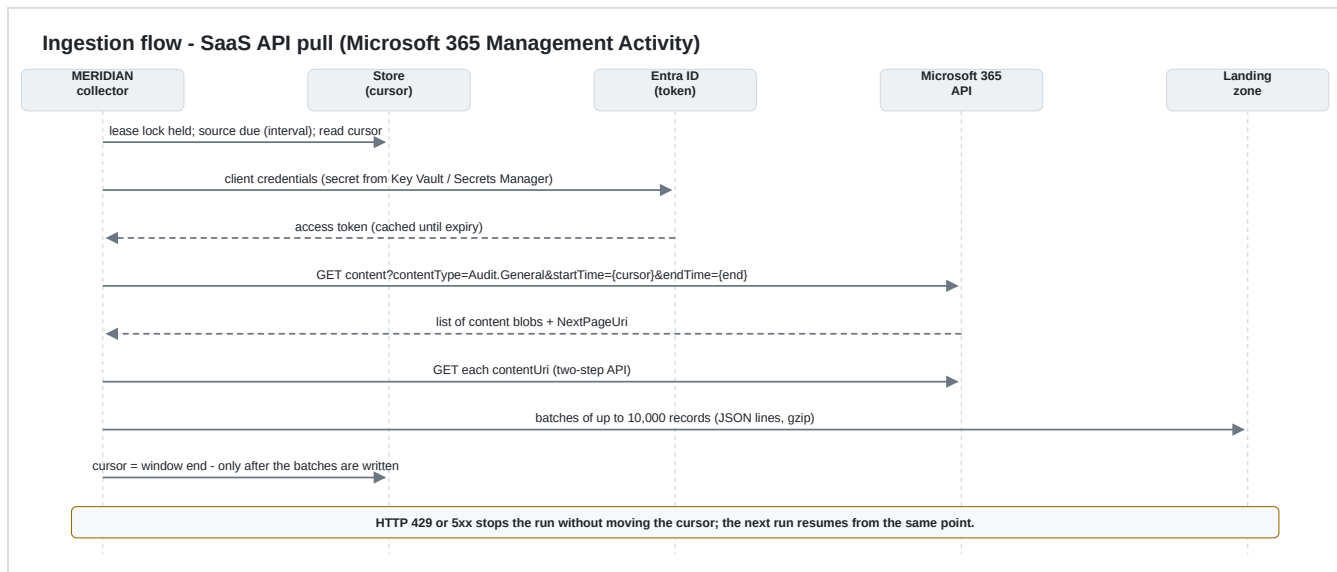


Figure 9.1 The API collector: the cursor advances only after the data is in the landing zone.

Signals

Metric	Alert when	Meaning
<code>meridian_source_last_batch_age_seconds{source}</code>	Greater than the source's normal interval $\times 3$, or -1	Silent source (PCI DSS 10.7: failure of a critical control)
<code>meridian_ingest_rejected_24h / meridian_ingest_events_24h</code>	Reject ratio above 1%, or rising	Format change at the sender, or a wrong format / timezone
<code>meridian_pull_last_run_ok{source}</code>	0 for two intervals	API credentials expired, throttling, API change
Dead-letter queue / poison queue depth	Greater than 0	A batch that cannot be processed
<code>meridian_work_items{kind="triage",status="queued"}</code>	Growing for 30 minutes	Detection backlog, not ingestion
<code>meridian_heartbeat_age_seconds{role="collector"}</code>	Greater than 300	Collector stopped

Troubleshooting

Symptom	Likely cause	Fix
Events land hours in the future or past	Naive local time without timezone, or a device clock problem	Set timezone on the source; fix NTP at the sender
Everything from a syslog source is class 0	The device's message format has no pattern	Add patterns (regex with named groups that are column names), then <code>meridian replay --prefix <source>/</code>
Windows events from Fluent Bit are class 0 or missing fields	<code>Event_Data_As_Map</code> not set	Set it; replay

Symptom	Likely cause	Fix
TLS handshake fails	Client certificate not from the configured CA, or SAN mismatch on the server certificate	Check <code>--tls-client-ca</code> and the certificate chain
HTTPS push returns 401	Wrong token for that source, or HMAC clock skew over 5 minutes	Each token is bound to one source; fix NTP
HTTPS push returns 413	Over 10 MB per request or over 64 MB decompressed	Reduce the shipper's batch size
API pull stuck	429 or 5xx from the API, or a cursor in the past beyond the API's limit	Check the logs; <code>meridian pull --source <key> --reset</code> re-reads from <code>start_minutes</code>
Duplicates after an API retry	Inclusive <code>since</code> at the boundary	Same <code>event_uid</code> ; de-duplicated at query time

10 Onboarding a source

Step	Action	Done when
01	Choose the path (section 3) and the source key. Record the owner and the expected daily volume	Entry in the source register
02	Take a sample, run <code>meridian map-test</code> , add <code>patterns / field_map / timezone</code> until the important columns are populated	Output filed; rejects 0
03	Add the source to the configuration (<code>sources:</code>); create its token, writer policy or pull secret	Deployed
04	Turn on the sender; confirm <code>meridian_source_last_batch_age_seconds</code> is fresh and rejects stay below 1%	24 hours clean
05	Check that the rules relying on it fire on a test event (for example a failed logon or <code>certutil -urlcache</code> in a lab)	Alert seen
06	Add the source to silent-source alerting and to the LODESTAR coverage list	Done

11 Security of the ingestion tier

Threat	Control
Spoofed or injected log data	Private network paths only; TLS with client certificates for syslog; HMAC or a source-bound token for push; IAM limited to one prefix for object storage. Telemetry is labelled untrusted for agents
One sender writing as another source	Tokens, writer policies and HMAC secrets are each scoped to a single source
Oversized or malicious bodies	Request and decompression caps (zip bombs → 413); 64 KB per syslog line; Windows XML with DTDs or entities is refused; one bad record never blocks a batch
Credential exposure	Secrets only in Key Vault / Secrets Manager; configuration refuses literal secrets; syslog keys held in memory
Tampering with stored logs	Landing immutable by convention and expiry only; lake under WORM; lake writer is the ingestion identity only, and the agents' identity has read access
Loss during outages	Source buffers, relay disk queues, receiver retry buffer, queue redelivery with a dead-letter queue, 90-day replayable landing

12 Limits and open items

Item	Position
UDP on Azure	Container Apps has no UDP ingress. Use a site relay or a VM-based receiver
Kafka	No native consumer. Bridge with Vector, Fluent Bit or Cribl to object storage or push
Dedicated mappers	Nine formats at release. Other products use json field maps, ocsf or patterns; dedicated mappers for high-volume sources are Phase 2 of the adoption plan
Fluent Bit positional inserts	Not mapped; Event_Data_As_Map is required
HEC acknowledgement	Not offered
Cross-batch duplicates	Kept, with the same event_uid; removed at query time
Lake compaction	Small files from low-volume sources raise query cost; compaction is Phase 1 work (EXECUTIVE_REVIEW \$5)

Sources for the external claims in this document

Source	What it establishes	Reference
Azure Monitor Agent: send data to Event Hubs and Storage	Preview; no new rules from February 2026; stops after 31 July 2026; Azure VMs only	learn.microsoft.com/azure/azure-monitor/agents/azure-monitor-agent-send-data-to-event-hubs-and-storage
NSG flow logs overview	NSG flow logs retire 30 September 2027; no new ones; migrate to VNet flow logs	learn.microsoft.com/azure/network-watcher/nsg-flow-logs-overview
Azure Container Apps ingress	HTTP and TCP ingress; no inbound UDP	learn.microsoft.com/azure/container-apps/ingress-how-to
Office 365 Management Activity API	Subscription start; content listing with NextPageUri; 24-hour windows within 7 days; content types	learn.microsoft.com/office/office-365-management-api/office-365-management-activity-api-reference
Fluent Bit winevtlog input	event_data_as_map, string_inserts, render_event_as_xml	docs.fluentbit.io/manual/data-pipeline/inputs/windows-event-log-winevtlog
NXLog Community Edition reference	im_msvistalog, om_ssl, om_tcp, xm_json, xm_syslog in CE; no om_http	docs.nxlog.co/ce/current
NXLog Agent om_http	HTTP(S) output with AddHeader, gzip, ndjson / jsonarray batching	docs.nxlog.co/agent/current/om/http.html
Winlogbeat outputs	Elasticsearch, Logstash, Kafka, Redis, file, console	elastic.co/docs/reference/beats/winlogbeat
PCI DSS v4.0	10.5.1: 12 months of audit history, 3 months immediately available	pcisecuritystandards.org
MERIDIAN repository	Every MERIDIAN behaviour stated here, with tests	meridian/ingest , meridian/mappers , tests/test_ingestion.py (v1.0.1)